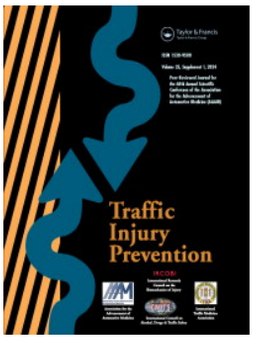

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ISSN: 1538-9588 (Print) 1538-957X (Online) Journal homepage: www.tandfonline.com/journals/gcpi20

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To cite this article: Angela Schubert, Eamon T. Campolettano, John M. Scanlon, Timothy L. McMurry & Thomas Unger (2024) Bridging the gap: Mechanistic-based cyclist injury risk curves using two decades of crash data, Traffic Injury Prevention, 25:sup1, S105-S115, DOI: 10.1080/15389588.2024.2400276

To link to this article: https://doi.org/10.1080/15389588.2024.2400276

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TRAFFIC INJURY PREVENTION
2024, VOL. 25, NO. S1, S105–S115
https://doi.org/10.1080/15389588.2024.2400276

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Bridging the gap: Mechanistic-based cyclist injury risk curves using two decades of crash data

Angela Schubert[a], Eamon T. Campolettano[b], John M. Scanlon[b], Timothy L. McMurry[b] and Thomas Unger[a]

[a]VUFO - Traffic Accident Research Institute at TU Dresden, Dresden, Germany; [b]Waymo LLC, Mountain View, California, USA

ABSTRACT

Objective: Injury risk curves are vital in quantifying the relative safety consequences of real-world collisions. Previous injury risk curves for bicycle-passenger vehicle crashes have predominantly focused on frontal impacts. This creates a gap in cyclist injury risk assessment for other geometric crash configurations. The goal of this study was to create an "omnidirectional" injury risk model, informed by known injury causing mechanisms, that is applicable to most geometric configurations.

Methods: We used data from years 1999–2022 of the German In-Depth Accident Study (GIDAS). We describe the pattern of injuries for cyclists involved in collisions with passenger vehicles, and we developed injury risk functions at various AIS levels for these collisions. A mechanistic-based approach accounting for biomechanically-relevant variables was used to select model parameters a priori. Cyclist age (including children) and sex were regarded as relevant predictors of injury risk. Speed and impact geometry were captured through a novel predictor, *Effective Collision Speed*, which transforms the vehicle and cyclist speeds into a single value and incorporates frictional considerations observed during side engagements. Cyclist engagement with the vehicle was captured with a variable demonstrating the potential for a normal projection. We additionally present analyses weighted toward German nationwide data.

Results: We identified 6,576 cyclists involved in collisions with passenger vehicles. AIS3+ cyclist injuries occurred most often in the head, thorax, and lower extremities. *Effective Collision Speed* was a strong predictor of injury risk. Collisions with a potential for a normal projection were associated with increased risk, though this was only significant at the MAIS2+F severity level. Younger children had slightly higher injury risk compared to young adults, while elderly cyclists had the highest risk of AIS3+ injury. Sex was a significant predictor only for the MAIS2+F injury risk curves.

Significance: U.S. cyclist fatalities increased 55% from 2010 to 2021. To reduce injuries and fatalities, it is crucial to understand cyclist injury risk. This study builds on previous analyses by including children, incorporating additional mechanistic predictors, broadening the scope of included crashes, and using weighting to generalize these estimates toward national German statistics.

ARTICLE HISTORY
Received 22 March 2024
Accepted 29 August 2024

KEYWORDS
Cyclist; bicycle; injury risk; logistic regression; GIDAS; passenger vehicles

Introduction

It is estimated that 100,000 cyclists are treated in hospital emergency departments each year in the United States for injuries following collisions with motor vehicles (Centers for Disease Control and Prevention 2023). With the growing popularity of cycling, not only for recreation but also for daily commuting, the number of cyclist fatalities has increased steadily in the U.S. (Buehler and Pucher 2021; Goel et al. 2021; NHTSA 2023).

Existing injury risk models

Injury risk curves play a vital role in automotive safety. They provide the quantitative bridge between a set of collision kinematics and its propensity to cause injury. This quantification has proven useful for a number of applications, including, among many others, implementation of advanced automatic crash notification (AACN), evaluating vehicle safety performance, designing safe roadways, and the evaluation of automated driving systems (ADS) (Kononen et al. 2011; Weaver et al. 2015; Jurewicz et al. 2016; Bareiss and Gabler 2020; Scanlon et al. 2021; Scanlon et al. 2022). Over the last decade, numerous researchers have developed injury risk models for collisions between cyclists and motor vehicles (Nie et al. 2013, Rosén 2013; Bahrololoom et al. 2020; Jeppson and Lubbe 2020; Lubbe et al. 2022). Most of these risk models exclusively consider collision events with the front of the vehicle, with only Bahrololoom et al. 2020 extending their model to additional crash configurations.

CONTACT John M. Scanlon johnscanlon@waymo.com Waymo LLC, 1600 Amphitheatre Pkwy, Mountain View, CA 94043, USA.
Associate Editor Sjaan Koppel oversaw the review of this article.
Supplemental data for this article can be accessed online at https://doi.org/10.1080/15389588.2024.2400276.
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While frontal contacts are the most common collision configuration among fatal events, non-frontal strikes still account for 9% of all cyclist traffic fatalities in the U.S. as of 2021 (NHTSA 2023).

All cyclist injury risk models published to date utilize some measure of collision impulse, most often represented by speed, as a primary predictor variable. Most commonly, vehicle speed at impact is the chosen value for modeling (Nie et al. 2013, Rosén 2013; Jeppson and Lubbe 2020). Recent work has begun to consider the influence of cyclist speed on injury risk (Bahrololoom et al. 2020; Lubbe et al. 2022; Macioszek and Granà 2021). For example, Lubbe et al. 2022 looked at frontal strikes only but considered a relative speed measure instead of absolute speed.

Bridging the gap: Cyclist injury mechanisms

Although these existing models may be appropriate for certain injury risk modeling applications, a holistic consideration of cyclist injury risk that reasonably covers most foreseeable configurations is vital for others. For example, consider ADS safety methodologies, which might include scenario-based testing or hazard analysis and risk assessment (ISO 26262 2018, (Webb et al. 2020; Kusano et al. 2022)). This study addressed this challenge by leveraging known injury mechanisms to model cyclist injury risk across reasonably foreseeable collision geometries. Known injury mechanisms are presented below, with their implementation for injury risk modeling discussed in the methodology section.

A cyclist involved in a collision with a passenger vehicle may become injured due to direct contact with the vehicle during *initial impact*, due to *secondary contact* with the ground, roadway environment, or other vehicles, due to *subsequent run over*, or due to some combination of these (Simms and Wood 2009; Badea-Romero and Lenard 2013; Gildea 2023).

In the initial impact phase, injury can occur as the cyclist engages with the striking vehicle's surfaces. This can involve the cyclist moving into various areas of the vehicle (e.g., hood, windshield, A-pillar). While there are multiple factors to consider during initial impact, ultimately, the injury risk is dependent on loading conditions and the body regions being engaged (Simms and Wood 2009).

During this initial impact phase, given the substantial mass differences between a vehicle and a cyclist, a maximal engagement collision would result in the cyclist traveling at the speed of the vehicle following initial impact; however, the cyclist may not always reach a common velocity with the vehicle during a collision. Neale et al. 2021 analyzed real-world pedestrian impacts and found clear dependency on projection distance based on whether the person was engaged with the frontal structures of the vehicle or with the side facing structures of the vehicles. While cyclists and pedestrians have some postural differences, their post-collision motion patterns are similar (Otte 1980; Otte 1989; Cardot et al. 2006; Simms and Wood 2009). Thus, one can imagine that, for contacts rearward of the A-pillar, the amount of longitudinal force from the vehicle imparted onto a cyclist would be less than had the cyclist engaged structures forward of the A-pillar that could work to more effectively push the cyclist in the direction of the vehicle. Consideration of the amount of potential engagement between the cyclist and the vehicle is critical for both the initial impact phase, as well as secondary contact risk, where the cyclist could be thrown.

During secondary contact, the cyclist may be projected from the vehicle onto other surfaces or objects. This can include falling to the ground and experiencing a combination of sliding, bouncing, and rolling before coming to rest. It can also include projection into other objects or vehicles. Like the initial impact phase, the injury outcomes depend on the loading conditions and body regions being engaged.

Both the initial impact and secondary contact phases have key dependence on the "projection" mechanism of the cyclist. There are generally two projection mechanisms: forward and wrap. A wrap projection is likely to occur when a cyclist is struck below their center of gravity, resulting in a motion differential between the upper and lower body, where the upper body wraps around vehicle structures and can lead to contact with the hood, windshield, or A-pillar. Conversely, a forward projection is likely to occur when the cyclist is struck near or above their center of gravity, such that the body is accelerated by the striking vehicle and projected forward away from the vehicle Simms and Wood 2009.

Lastly, any event in which the cyclist falls to the ground has the potential to result in subsequent run over. This risk is highly dependent on surrounding traffic, awareness of the impacting driver, impact configuration, and the projection mechanism. Given these highly variable dependencies, and the fact that run over events are more closely associated with a crush mechanism than an impact mechanism, injury outcomes in these events are less closely tied to speed. As part of the overall crash sequence, including cases which involved subsequent run over would likely result in a conservative risk assessment. It is important to note that events which exclusively involve a vehicle running over a cyclist (i.e., previously fallen cyclist in the roadway) should not be assessed in the same manner as impact events.

Research objectives

This study provides key step change improvements to the existing literature by considering non-frontal collisions and introducing a set of biomechanics-based predictors for injury risk modeling. Chief among these are defining a novel "*Effective Collision Speed*" term that incorporates a physically relevant effect of impact location on modeled speed and accounting for geometric factors that influence whether a normal projection is likely to occur, which extends the concept of a forward projection in frontal collisions to all impact sides. Two specific research questions were posed: (1) what are the general injury risk patterns for cyclists involved in collisions with passenger vehicles? And (2) how does cyclist injury risk depend on models that leverage these novel predictors?

Methods

Data source

This study relies on motor-vehicle-cyclist collisions captured as part of the German In-Depth Accident Study (GIDAS). GIDAS is a joint project of the German Federal Highway Research Institute (BASt) and the German Association for Research in Automobile Technology (FAT) (German In-Depth Accident Study 2024). GIDAS has collected highly detailed data on approximately 2,000 traffic crashes per year since 1999 in Dresden, Hanover, and their surroundings. We used the GIDAS database as of December 31, 2022; which included 41,595 fully documented and reconstructed crashes (FAT et al. 2022).

To be included in GIDAS, at least one person involved in the crash must have been injured. The data is collected directly at the crash scene and includes information vital to performing a crash reconstruction, participant interviews, and detailed documentation of all injuries including medical records. Each crash in GIDAS is then reconstructed from the first critical situation to the final position of those involved. Approximately 3,500 individual features are documented for each crash. GIDAS provides representative information on the occurrence of injury crashes in Germany as the survey areas are representative of Germany's typical topography. The survey follows a precisely defined sampling plan, and the number of cases is correspondingly high.

For this analysis, we extracted cases where a bicycle had collided with exactly one passenger vehicle and applied exclusion criteria as outlined in Table 1.

To be comparable to the U.S. categorization of passenger vehicles (NHTSA 2023), we defined passenger vehicles as vehicles weighing less than 10,000 lbs in the M1 and M2 passenger vehicle or N1 smaller goods vehicle classes according to the official ECE definition UNECE 2005 (Table A1).

Injury severity was measured using the person-level Maximum Abbreviated Injury Scale (MAIS, 2015 revision of AIS), which represents the highest injury level taken across all the cyclist's documented injuries (AAAM 2016).

Injury risk model parameters

We modeled the complex biomechanical interaction in bicycle-passenger vehicle collisions by incorporating important influences into an injury risk function. In our models, injury severity was classified using cyclist-level MAIS (AAAM 2016). Relevant parameters were identified a priori, and statistical model selection was not performed. Specifically, we selected *Effective Collision Speed*, age, sex, and a variable differentiating between the expected cyclist projection mode as our predictor variables.

Table 1. Exclusion criteria to extract cases from the GIDAS dataset.

Exclusion criteria
Bicycles with trailers
Multiple persons on a bicycle
Bicycle collisions with vehicle trailers
Cyclist fell before impact with the car
Cyclists who experienced subsequent collisions with other road users (but no exclusion of cases with multiple collision with the same passenger vehicle)
Cases with unknown cyclist age, injury, and collision impact locations on the bicycle and car
Cases with unknown collision speeds and collision angle to determine the relative speed between bicycle and passenger vehicle
Exclude cases with passenger vehicle slip angles greater than 20°, exclude skidding

Model form and outcome variables

Injury risk was modeled using binary logistic regression with injury outcome represented as a binary variable based on the MAIS level or a recorded fatality (e.g., MAIS2+F indicates that the cyclist sustained at least one injury coded as AIS2 or greater; or was recorded as fatally injured, irrespective of the MAIS level). We created risk curves for MAIS2+F, 3+F, 4+F, and 5+F levels.

Modeling Effective Collision Speed

As mentioned, this study aimed to implement an *Effective Collision Speed* term. This section presents the computation of this term in two parts. First, we present the variables available in GIDAS and simple derivations of various velocity components. Second, we derive our *Effective Collision Speed* term from these terms.

GIDAS speed variables. Both the vehicle speed and cyclist speed at impact are defined variables *(VK* variable in GIDAS). The variable *KWINK* describes the relative angle between each computed *VK* (Figure A1), which can be used to transform the velocity of the cyclist into the vehicle reference frame. Eq. (3) shows the absolute magnitude of the relative speeds derived from the relative speeds in lateral and longitudinal direction.

$$v_{lateral} = VK_{bicycle} \cdot \sin\left(KWINK\right) \tag{1}$$

$$v_{longitudinal} = VK_{passenger\ vehicle} - VK_{bicycle} \cdot \cos\left(KWINK\right) \tag{2}$$

$$v_{rel} = \sqrt{v_{lateral}^2 + v_{longitudinal}^2} \tag{3}$$

Effective Collision Speed: The primary model predictor. Our computation of *Effective Collision Speed* independently considers the likely impulse experienced by the rider during (a) initial impact and (b) subsequent contact. These phases independently contribute to injury risk. There is, of course, the potential for a subsequent contact to also exacerbate the injuries from some initial impact. As the effect of vehicle and cyclist speed along with collision geometry influences these impulses, we sought to model the physical phenomena as outlined in the introduction. For instance, imagine a scenario where a cyclist traveling at 15 kph perpendicularly strikes the side of a stopped vehicle. Were that contact to occur rearward of the A-pillar, the cyclist's forward momentum is effectively stopped by the side vehicle geometry, and the cyclist speed (and relative speed) is representative of the impulse associated

with the initial contact. Conversely, were the contact to occur forward of the A-pillar, the cyclist may separate from their bicycle and be thrown at 15 kph as the front vehicle structures arrest the forward motion of the bicycle. In this scenario, the cyclist speed is representative of the impulse associated with the cyclist being thrown and experiencing a subsequent contact.

Across all crash configurations, the goal of the *Effective Collision Speed* term was to effectively capture the impulse associated with the worst-case single impact event a cyclist may experience. This was achieved by considering the maximum of cyclist speed ($VK_{bicycle}$), friction-adjusted vehicle speed ($v_{passenger\ vehicle-fric}$), and friction-adjusted relative speed ($v_{rel-fric}$):

$$Effective\ Collision\ Speed = \max\left(v_{passenger\ vehicle-fric}, VK_{bicycle}, v_{rel-fric}\right) \tag{4}$$

In this computation of *Effective Collision Speed*, we first defined a friction adjustment term, A. It only applies to the velocity component in the direction of the vehicle's travel (longitudinal velocity components), capturing that the cyclist may not fully engage with the vehicle during the contact, and relative sliding may occur, resulting in reduced energy transfer to the cyclist. For all frontal and rear contacts, no frictional effect is modeled (i.e., $A = 1$), which physically treats these strikes as causing the struck cyclist to adopt the speed of the vehicle during impact; this approach was also used for side contacts forward of the A-pillar. In these contacts, the potential for the cyclist to contact outer vehicle structures like the front fender or the articulating front wheels is present, with subsequent wrapping around the vehicle and potential interaction with the A-pillar, windshield, or other parts of the car that impart forward momentum. Additionally, dooring events were also modeled as a non-sliding engagement ($A = 1$). Therefore, sliding contacts ($A < 1$) were modeled only for side contacts rearward of the A-pillar, where partial engagement between cyclist and vehicle is expected, as outlined in the introduction. Throughout the manuscript, "front contact" refers to all contacts forward of the vehicle A-pillar, while "side contact" refers to all other side contacts. This study evaluated and selected a friction factor (A) that best fit the data. This essentially solves for the most representative friction factor for cyclists engaging the side of the vehicle. The physical interpretation of the friction factor is that side collisions are associated with a (1-A) reduction in imparted impulse from the vehicle relative to a full engagement collision.

The friction-adjusted relative speed is proportional to the expected momentum transfer and impulse imparted on the cyclist by the vehicle during contact. To reiterate, friction adjustment was only applied to forces in the longitudinal direction. This implies that the cyclist's full momentum in the lateral direction (again, in the vehicle reference frame) is being modeled as being completely halted. This decision was made to best reflect that there is likely additional injury risk associated with increasing cyclist speed, even in situations where cyclist speed may not dominate the risk assessment. For example, consider two cyclists being struck in the side by a forward-traveling vehicle (15 kph). One is struck while at a standstill, and the other is traveling at 10 kph prior to impact. While most of the injury potential is associated with the vehicle's speed, there is additional risk for the cyclist traveling at 10 kph that should be considered. The friction-adjusted relative speed achieves this. Shown in Equation (6), the longitudinal relative velocity term ($v_{longitudinal}$) was transformed using the friction factor (A) to form the friction-adjusted longitudinal relative velocity term ($v_{long-fric}$).

$$v_{rel-fri} = \sqrt{v_{lateral}^2 + v_{long-fric}^2} \tag{5}$$

$$v_{long-fric} = v_{longitudinal} \cdot A \tag{6}$$

The friction-adjusted vehicle speed ($v_{passenger\ vehicle-fric}$) term primarily represents a potential permutation where some amount of longitudinal vehicle momentum is imparted onto the cyclist during the engagement. Depending on the crash configuration, this may represent the impulse associated with initial impact or the resultant speed available for a subsequent collision event. For frontal or rear contacts where the geometry effectively induces full engagement ($A = 1$), no reduced engagement is modeled. As outlined in Equation (7), during side contacts, where there is some relative sliding between the cyclist and the vehicle, there is necessarily a decreased impulse from the vehicle imparted onto the cyclist. For example, consider a cyclist traveling at 10 kph alongside a vehicle traveling at 20 kph (relative angle of 0 degrees), and the side of the vehicle contacts the side of the cyclist. Using either cyclist speed or closing speed would estimate injury risk based on 10 kph. Using absolute vehicle speed would overestimate risk at 20 kph, given that there is not expected to be full engagement between the vehicle and the cyclist. The friction-adjusted vehicle speed term captures this decreased energy transfer during the collision event but also represents the dominant term in the evaluation of injury risk, as the cyclist could be accelerated up to a higher speed as a result of the contact.

$$v_{passenger\ vehicle-fric} = A \cdot VK_{passenger\ vehicle} \tag{7}$$

Modeling potential normal projection

As discussed in the introduction, beyond collision speed, the manner in which the cyclist rider engages with the vehicle, through either a wrap or frontal projection, influences injury risk (Ackery et al. 2012; Edwards and Leonard 2022; Macioszek and Granà 2021; Monfort and Mueller 2023). To capture that cyclist motion following initial impact can vary based on the geometric engagement with the vehicle, as well as extending our model beyond application to frontal collisions, we developed a variable, NPROJ.

As GIDAS does not encode all geometric characteristics regarding the impacted passenger vehicle structures, collision configurations were assigned as either being a potential normal projection or not based on the orientation of the bicycle

at impact and whether a tall, flat portion of the vehicle was struck (Table 2) (identified *via* the GIDAS variables VDI2, FZGKLASS, ABF, FZART).

Other model predictors

Age is a known influential covariate for injury prediction (Lubbe et al. 2022; Macioszek and Granà 2021). Children are often excluded from injury risk analyses, but, given that approximately 15% of the cyclists in the data are under 18 years old, we felt it was important they be included.

Building upon the methodology outlined in Schubert et al. 2023; we utilized a natural cubic spline to model age. This approach yielded two terms – within the logistic regression model (*Age Spline 1* and *2*) and follows the hypothesis that young adults exhibit greater resilience to injury compared to both younger children and older adults (Forman and McMurry 2018; Forman and McMurry 2018). Analogous to Schubert et al. 2023; a sensitivity analysis comparing the spline model with a model using age categories found good agreement between the models (Figure A2), and the spline model was selected due to its continuous representation.

Sex-specific differences in injury risk were also investigated (male: $SEX = 0$; female: $SEX = 1$).

In initial evaluation, we considered cyclist helmet use, but chose to not include it for two reasons. First, few of the cyclists in our data set (17.4%) were wearing helmets, and second helmets only provide some protection against a relatively small proportion of potential injuries (i.e., head injuries). In preliminary analyses, the unadjusted odds ratio of sustaining an AIS3+ injury overall was approximately 1 (i.e., no effect of helmet usage), whereas performing the same calculation and only considering head injuries indicated that unhelmeted cyclists were 5.5x more likely to sustain an AIS3+ head injury than their helmeted counterparts (Table A2). Thus, the data indicate that there is a clear effect of helmet usage on head injuries, but not on the overall MAIS level.

Model evaluation

The predictors were chosen based on their biomechanical significance and relevance to crashes. After creating the model, for completeness we considered the areas under the receiver operating characteristic curve (AROC) and the precision-recall curve (APRC) for a 5-fold cross-validation.

We conducted a comparative analysis between the unweighted models and models re-weighted to reflect the entire injurious crashes of Germany. Reweighting helps to account for a slight overrepresentation of serious and fatal collisions in GIDAS (Pfeiffer and Schmidt 2006). The re-weighting process involved applying weighting factors derived from the 2022 German national data (DESTATIS 2022). The Federal Statistical Office of Germany provided distributions specific to bicycle-related crashes. Normalized weighting factors were applied as in Schubert et al. 2023. Further details can be found in the Appendix (supplementary material).

Table 2. Definition of the predictor NPROJ.

Potential normal projection – *NPROJ* = 1	No potential normal projection – *NPROJ* = 0
• Front of a taller vehicle (e.g., a van) strikes a bicycle. • Bicycle has a frontal impact with the tall portion of the vehicle (e.g., the front wheel of the bicycle strikes the passenger compartment of a sedan, or any part of a van/SUV/hatchback/wagon behind the A-pillar). • Rear of a tall vehicle strikes a bicycle while reversing. • Dooring.	• Front of a small vehicle is hit (front defined until A-pillar). • Side or rear of the bicycle has contact behind the A-pillar of the car with the side of the car. • Bicycle hits the rear of a sedan. • Side or rear of the bicycle has contact with the rear of a non-reversing tall vehicle.

Results

Descriptive analyses

After applying our exclusion criteria to GIDAS, the dataset consisted of 6,576 bicycle-to-passenger vehicle collisions. While a cyclist being struck in the side by the front of a forward-traveling vehicle is the most common impact configuration, occurring in 59% of cases, other collision scenarios occur at non-negligible frequencies (e.g., collision between the front of the bicycle and the front of the vehicle observed in 18% of cases) that bear consideration in the evaluation of generalized cyclist injury risk (Figure 1).

From the passenger vehicle perspective, impacts with the vehicle front were associated with higher *Effective Collision Speeds* on average than for all other configurations, as well as having a wider range of observed speeds (Figure 2). From the cyclist perspective, cases where the rear of the bicycle was struck were associated with higher speeds than other impact locations even though the observed frequency of these events was much lower (Figure 1 and Figure 2).

Median *Effective Collision Speed* was approximately 21 kph, with a slightly lower median value for collisions in which the cyclist experienced a potential normal projection (Figure A3). Events with no potential normal projection were observed more than 4x as often as potential normal projection contacts.

Cyclists in this dataset ranged in age from 3 years old to 92 years old, with a median age of 36 years old (Figure A4). The largest concentration of cyclists was between the ages of 20 and 35. The age distributions by sex differed slightly with a higher proportion of females older than 55 years and a higher proportion of males younger than 15 years. Overall, approximately 42% of cyclists in this dataset were female.

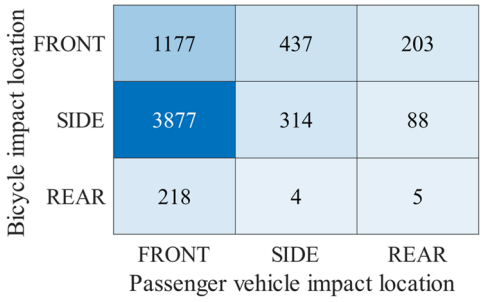

Figure 1. Number of crashes depending on passenger vehicle and bicycle impaction location – FRONT for passenger vehicles defined as impact forward of the A-pillar - excluding dooring cases ($n = 253$) given the unique characteristics associated with that specific impact configuration.

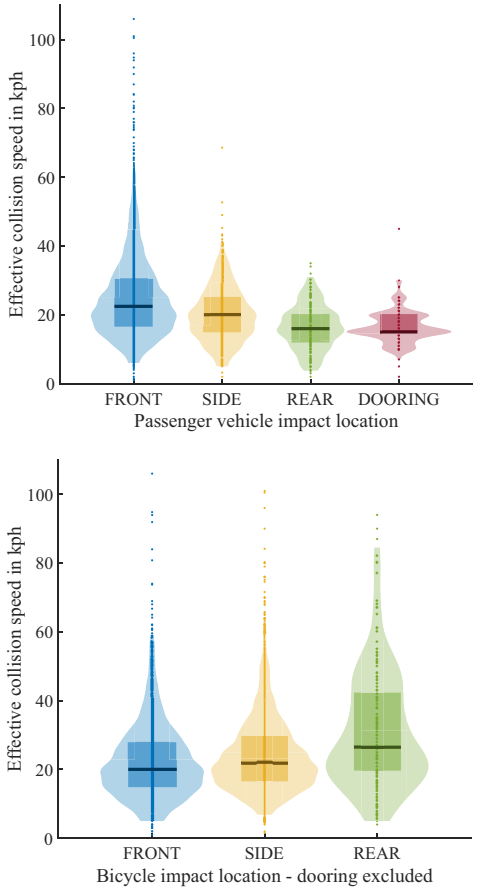

Figure 2. Distribution of the *Effective Collision Speed* depending on the impact direction - FRONT for passenger vehicles defined as impact forward of the A-pillar. Showing the kernel density and a boxplot with the median as solid line and the quartiles.

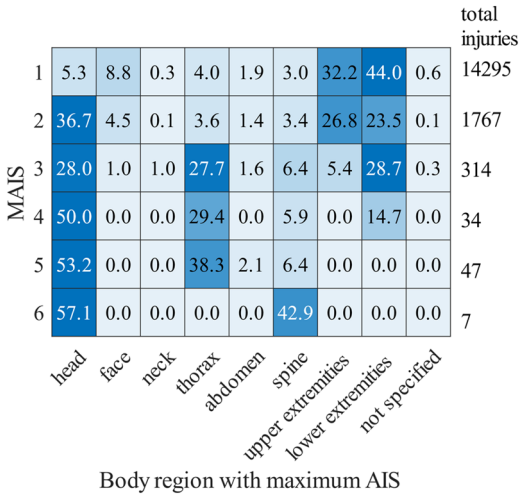

Figure 3. Body region distribution (%) by cyclist MAIS level for cyclists' most severe injury(ies) (all injuries at each cyclist's overall MAIS level are included).

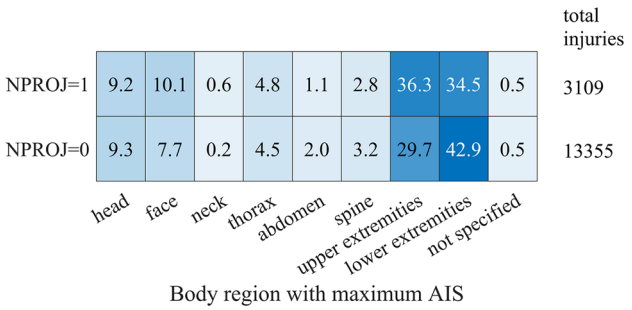

Figure 4. Body region injury distribution (%) by potential normal projection for injuries with the cyclists' maximum AIS values.

Injury patterns

We describe common cyclist injury patterns by tabulating the body region(s) where each cyclist's maximum AIS-level injury occurred (Figure 3). Serious or worse injuries (AIS3+) were most frequently observed in the head, thorax, and lower extremities, with head injuries comprising 33% of the AIS3+ injuries. Brain hemorrhages (43%) and skull fracture (30%) represented the most common AIS3+ head injuries. Rib cage/sternal fractures (70%) and lung injury (25%), i.e., contusion, laceration, hemo- and/or pneumothorax, represented nearly all AIS3+ thorax injuries. Femur fractures (61%) were the most common AIS3+ lower extremity injuries, followed by tibia fractures (25%) and pelvis fractures (12%). Among the small sample of dooring events, 63% of AIS3+ injuries were head injuries (Figure A5).

We also investigated differences in injury patterns due to potential cyclist engagement with the vehicle as captured by *NPROJ*. Collisions with potential normal projection were associated with nearly the same distributions of injuries by MAIS severity level as collisions with no potential normal projection despite lower *Effective Collision Speeds* (Figure A3) for potential normal projection. Collisions with no potential normal projection were most commonly associated with injuries to the lower extremities, representing approximately 43% of all injuries at the maximum AIS level (Figure 4). While collisions with potential normal projection were less commonly observed, there was a noted increase in the proportion of injuries to the upper extremities (36.3% vs. 29.7%) and face (10.1% vs. 7.7%), in comparison to collisions without normal projection.

We evaluated the GIDAS-attributed injury causing parts for serious or greater (MAIS3+) severity injuries (Figure 5). Outer vehicle structures cause a higher proportion of injuries in cases with no potential normal projection (38.1%) compared to potential normal projection events (28.2%), while the opposite is true for contacts with the road surface, vehicle underside, or multiple combinations (more than one injury causing part may be coded).

Estimation of friction factor

We estimated the value of the friction factor by a manual approximation of maximum likelihood (the same approach used in logistic regression to estimate the model coefficients). In detail, we fit MAIS2+F and MAIS3+F injury risk models with a single predictor variable, *Effective Collision Speed*. The friction factor was varied from 0 (no friction), where vehicle speed does not contribute to risk in a side contact, to 1, where a side contact is modeled similarly to a frontal contact, in increments of 0.05. The maximum log-likelihood for both the MAIS2+F and MAIS3+F models

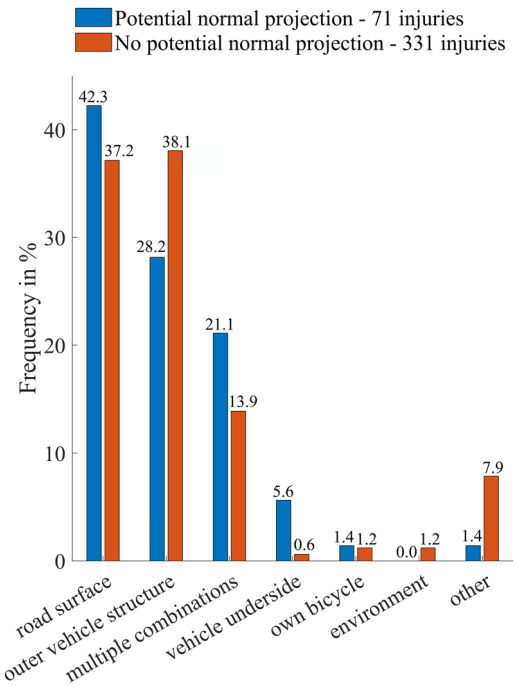

Figure 5. Distribution of injury causing parts for all MAIS-level AIS3+ injuries.

was observed at $A = 0.7$, and this value was selected as the optimized parameter for computing *Effective Collision Speed* in the remainder of the analyses. Assuming a 90-degree collision angle, this means that the contribution of vehicle speed to injury risk would be decreased by 30% for side contacts relative to a frontal contact (e.g., the vehicle speed component for a vehicle traveling at 20 kph and being struck by a cyclist in the side would be 14 kph).

Further, *Effective Collision Speed* was compared to relative speed (VREL) and collision speed (VK) of the passenger vehicle and found to produce noticeably higher log-likelihoods (Figure A6).

Table 3. Data sample after excluding unknowns of independent variables.

Total – 6,576		
	Count	*%*
MAIS2+F	1,644	25.0
MAIS3+F	302	4.6
MAIS4+F	71	1.1
MAIS5+F	45	0.7

Injury risk curves

Serious or greater (MAIS3+F) injury outcomes represented fewer than 5% of the dataset used to model injury risk (Table 3). All fatalities in the dataset had an AIS code of 3 or higher.

Given the larger injury counts, presentation of injury risk models will focus on the MAIS2+F and MAIS3+F models (Table 4). Results pertaining to the MAIS4+F and MAIS5+F models are included in the Appendix (Table A4) and will not be discussed in detail because the limited number of MAIS4+F and 5+F injuries limit their statistical power. The AROC and APRC of all models are shown in the Appendix (Table A3).

As has been reported previously, speed, here modeled as *Effective Collision Speed*, and age have the largest effect on injury risk, regardless of injury severity level (Table 4). More energy at the time of impact, i.e., higher *Effective Collision Speed,* was associated with higher injury risk (Figure 6). The effect of the age spline necessitates both age terms being considered in conjunction to evaluate the age effect. Elderly cyclists were observed to be most vulnerable relative to adults, while younger children exhibit higher injury risk to a lesser extent (Figure A9). Given the relative paucity of children in the dataset, the confidence bounds are necessarily wider than for other population groups.

Sex was observed to be a statistically significant predictor at the MAIS2+F severity level, with females exhibiting a higher injury potential than males. This effect was not observed for serious or greater (MAIS3+F) injuries (Figure A8).

Collisions in which there was a potential for a normal projection were associated with increased risk overall. This effect was statistically significant at the MAIS2+F severity level but not the MAIS3+F level (Figure A7).

Risk curves at the MAIS2+F and MAIS3+F injury severity levels were also developed using the re-weighted data (Table 4). Similar trends and effects for all predictors were noted when compared to the unweighted models. Weighting of the dataset toward national German statistics resulted in lower injury risk estimates than the unweighted dataset, which is expected given the GIDAS sampling bias toward collisions involving serious or greater (MAIS3+F) injuries (Figure 7).

Table 4. Cyclist injury risk models for collisions with passenger vehicles for unweighted and weighted dataset.

		Unweighted			*Weighted*		
		Estimate	*Standard Error*	*p-value*	*Estimate*	*Standard Error*	*p-value*
MAIS2+F	*Intercept*	−2.867	0.151	0.000	−3.197	0.172	0.000
	Agespline1	0.026	0.248	0.000	0.364	0.291	0.000
	Agespline2	0.204	0.075		0.274	0.088	
	SEX Female	0.193	0.061	0.002	0.178	0.067	0.008
	NPROJ	0.210	0.078	0.007	0.236	0.083	0.004
	Effective Collision Speed	0.051	0.003	0.000	0.046	0.003	0.000
MAIS3+F	*Intercept*	−5.730	0.381	0.000	−5.903	0.459	0.000
	Agespline1	−0.147	0.656	0.000	−0.525	0.813	0.000
	Agespline2	0.364	0.201		0.215	0.248	
	SEX Female	−0.018	0.129	0.888	0.069	0.164	0.676
	NPROJ	0.266	0.168	0.114	0.345	0.201	0.085
	Effective Collision Speed	0.063	0.004	0.000	0.059	0.004	0.000

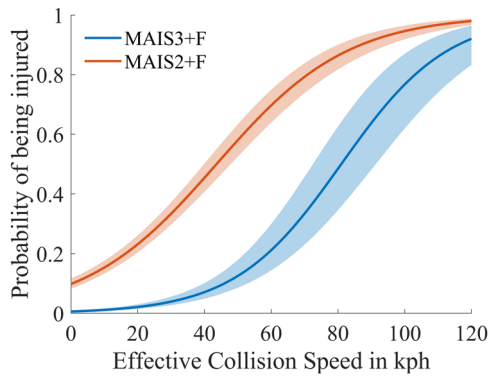

Figure 6. Injury risk of being MAIS2+F or MAIS3+F injured with confidence intervals for NPROJ = 1, age = 36 years (median age of cyclists in dataset), SEX = female.

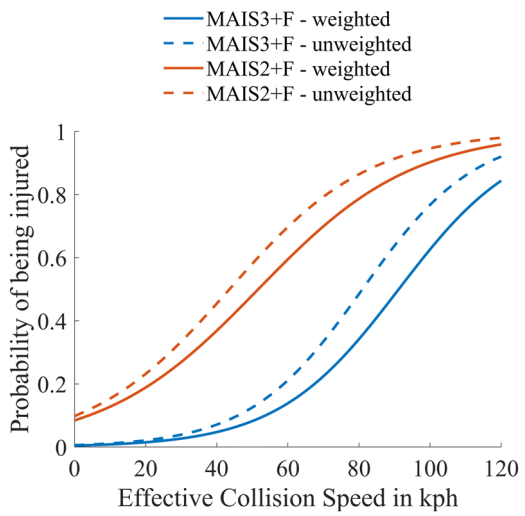

Figure 7. Comparison of risk curves for original and weighted dataset for NPROJ = 1, age = 36 years (median age of cyclists in dataset), SEX = female.

Discussion

This work builds on existing cyclist injury risk models by presenting injury patterns for common collision configurations and extending application to non-frontal collision events. The resulting risk models allow for "omni-directional" assessment of risk and capture biomechanically-relevant predictors. Using a unique speed parameter as the primary predictor variable (discussed below), this model also considered cyclist age and sex, two well-established biomechanical factors related to human injury tolerance. Lastly, a parameter (*NPROJ*) capturing the propensity for a collision to result in a normal projection away from the vehicle following initial contact was included. This factor ensures that we appropriately capture the increase in risk associated with potential subsequent contacts. All parameters were selected *a priori*.

"Effective Collision Speed" as an improved primary predictor

This is the first study on cyclist injury risk in collisions with passenger vehicles to consider all impact directions. Previous research has largely centered on contacts with the front of the vehicle, which does capture the majority of collision events involving cyclists. We found only one paper developing risk curve from non-frontal impacts (Bahrololoom et al. 2020). However, non-frontal contacts comprise nearly 25% of events involving AIS3+ injuries within our dataset. Further, among cases involving injury at the MAIS3+ severity level, more than 60% involved injuries induced by contact beyond the initial contact with the vehicle (Figure 5). Taken together, these highlight the importance of considering the different phases of impact for cyclists and for extending model applications beyond frontal collisions.

Ultimately, the utility of this model and *Effective Collision Speed* become apparent when considering other collision geometries. Prior to this study, the most commonly-used speed predictors were absolute vehicle speed or relative speed. While these approaches serve as good predictors of injury risk for the most common configuration of the front of a vehicle striking the side of a cyclist, these measures are not generalizable to other potentially injurious crash configurations. Consider a cyclist traveling at 15 kph in front of a vehicle traveling at 20 kph. The trailing vehicle contacts the cyclist from behind. Were we only to consider closing speed, we would model this event using a speed input of 5 kph. This closing speed analysis ignores the effect of travel speed, where the cyclist could fall at their initial travel speed (15 kph) or be projected at the vehicle's travel speed (20 mph) – making subsequent contact with the ground or other objects. This study considered the maximum potential impulse among these potential mechanistic permutations. In this way, the model, when applied in practice, will be less prone to risk underestimation.

Similarly, consider a parallel engagement (i.e., relative angle of 0 degrees) in which a cyclist traveling at 20 kph is contacted on their side by the side of a vehicle traveling at 10 kph. Using only vehicle speed or closing speed would fail to capture the associated risk of the cyclist losing control of their bicycle at 20 kph and experiencing a fall to the ground or contact with another object in the roadway environment. Again, failure to consider this potentially highest risk component of the impact could lead to an underestimation of the true risk of that engagement.

The most recent research on cyclist injury risk Lubbe et al. 2022 also leveraged GIDAS data, but focused on frontal impacts, using closing speed as the speed parameter to predict injury risk. While these differences preclude an exact comparison with the models developed in this study, some relative evaluation is still attainable by treating our speed predictor, *Effective Collision Speed,* and the closing speed value in the Lubbe et al. 2022 models as equivalent. From the lens of a common injury risk estimate, 10% MAIS3+F risk is associated with a closing speed of 46 kph in the Lubbe et al. 2022 model, compared to an *Effective Collision Speed* of 54 kph for our weighted model. Incidentally, our unweighted model predicts 10% MAIS3+F risk for an *Effective Collision Speed* of 46 kph.

Friction-adjusted speed

The adjustments to vehicle speed to capture reduced energy transfer during certain collision configurations represent a novel methodology that is ultimately consistent with the observed physical outcomes. It has previously been shown that pedestrians and cyclists have similar post-collision motion patterns and

that pedestrians experience reduced energy transfer during impact with vehicles for side contacts relative to frontal strikes (Otte 1980; Otte 1989; Simms and Wood 2009; Cardot et al. 2006; Neale et al. 2021). As determined through optimization in this study, these side engagements are associated with 30% less impulse transfer than full engagement contacts. This is not surprising when one considers the physics of these side contacts, where (1) relative sliding between the bicycle and the vehicle may occur during contact; or (2) deformation of the bicycle and the presence of taller side vehicle structures may dampen the effect of vehicle speed relative to other factors.

Excluding helmet status from risk models

As noted in the methods, helmet use was not modeled in the injury risk functions developed in this study. This was primarily done for two reasons. First, helmet use in this GIDAS sample was only observed in approximately 17% of all cases. Secondly, the effect of helmet use is only relevant for head injuries and/or instances where the cyclist's head is impacted. For example, a helmeted cyclist who sustained an open tibia fracture (AIS3 injury) would be counted as sustaining an injury at the MAIS3+ severity level while wearing a helmet, when the helmet use is unrelated to the sustained injury. As the injury risk models developed in this study assessed generalized whole body injury risk, it did not seem appropriate to model helmet usage. Further, the ability of a bicycle helmet to attenuate impact forces and mitigate injury potential has been reported to vary considerably by model (Mills and Gilchrist 2008; Stigson et al. 2017; Bland et al. 2018; M. Bland et al. 2020). That being said, it should be noted that more than 2/3 of all injuries at the MAIS4+ severity level were head injuries, and the head injury-specific odds ratios (Table A2) clearly support the protective capacity of helmets, as has been reported elsewhere (Olivier and Creighton 2017; Hoye 2018, Büth et al. 2023; Campolettano et al. 2023).

The most recent national data from the United States on cyclist attitudes and behaviors, including helmet usage, resulted from a 2012 National Highway Traffic Safety Administration survey, which reported that only 28% of cyclists always wore a helmet, with nearly half of all respondents (46%) reporting they never wore a helmet (Schroeder and Wilbur 2013). These results were similar to what was reported from a consumer–based survey carried out the same year, where 29% of cyclists always wore their helmet and 54% never did (Jewett et al. 2016). Compared to Germany, these numbers are actually slightly higher. In 2016; the German Federal Highway Research Institute reported that only 17% of cyclists wore helmets on urban roads (German Federal Highway Research Institute 2017), though this number increased considerably in the last years to 40.3% in 2022 (German Federal Highway Research Institute 2023). It is important to note that the distribution of injuries may differ in the United States and over time as a result of these differences in helmet use.

Limitations

There are several limitations of the underlying data within GIDAS. First, it is important to note that GIDAS covers crash data from more than two decades, which also means that any improvements made to frontal structures of passenger cars in relation to vulnerable road users are not reflected in the models. This is particularly relevant given that Euro NCAP began incorporating VRU-focused safety assessments as part of their safety ratings and has iterated on these over the years. Further, GIDAS' sampling criteria requires suspected injury to at least one party in order for the case to be investigated. This necessarily biases the database toward higher severity collisions. This is best seen by the logistic risk curves predicting nonzero risk for *Effective Collision Speeds* of 0 kph. As a result, the models err toward conservatism in the evaluation of injury risk due to this potential bias.

Further, limitations associated with modeling decisions cannot be overlooked. The *Effective Collision Speed* term developed in this study represents an approach to assessing risk that aimed to capture physically relevant phenomena regarding differences in level of engagement for side contacts. While there was a clearly observed statistical relationship as captured by the friction factor, A, this may not be the only way of capturing this effect. Given our consideration of the foreseeable crash types observed in GIDAS, we believe this model, when used to evaluate individual cases, will produce generally conservative risk estimates.

As Monfort and Mueller 2023 states "the size and shape of SUV front ends are responsible for the differences in bicyclist injury outcome." Since we had no information about the heights of the hood leading edges and the center of gravity of the cyclist, we made the simplification to classify vans as taller vehicles concerning the vehicle front for defining the variable *NPROJ*. We did not include SUVs in this definition for front impacts due to this missing information if the cyclist was struck above their center of gravity. Additionally, the segment of SUVs covers a wide range of different car sizes, and the German vehicle fleet is different from that in the U.S., where larger vehicle models are common.

Notable differences exist between the German and U.S. populations that must be considered. The aforementioned differences in vehicle fleet composition between the two countries may lead to some unquantified accuracy deviations when applying these models for use in the U.S. Also, differences in cyclist behavior (e.g., riding mileage, helmet usage) and roadway infrastructure (i.e., presence of dedicated cyclist lanes) may also affect injury outcomes. According to Goel et al. 2021; the mode share of cycling in the U.S. is 1.1%, while in Germany it is 9.3%. Additionally, the percentage of females participating in cycle trips is 49.2% in Germany, compared to only 30.2% in the U.S. Despite these limitations, GIDAS represents the best-available source of field crash data for VRU collision events.

Conclusion

This study provided a needed paradigm shift in cyclist injury risk assessment. By considering non-frontal impacts and the various geometric-dependent mechanisms affecting injury risk, the study generated injury risk curves that captured variable risk due to cyclist and vehicle kinematics and projection type (potential normal projection). Most importantly, unlike models dependent on vehicle speed or relative impact speed

alone, the formulation of this model reduces the propensity for underestimating injury risk. Furthermore, we seek to enhance and broaden the understanding of the various injury patterns resulting from those crashes. We believe these results are particularly important for designers and evaluators considering the use of injury-risk based metrics in ADS evaluation on a case-by-case basis or in-aggregate, where robust, conservative measures are needed for safety determination.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

Waymo LLC, 1600 Amphitheater Pkwy, Mountain View, CA 94043, US has commissioned the VUFO – Verkehrsunfallforschung an der TU Dresden GmbH (Traffic Accident Research Institute at TU Dresden) to work on the mentioned research project.

ORCID

Angela Schubert http://orcid.org/0009-0001-9639-3149
Eamon T. Campolettano http://orcid.org/0000-0002-0540-1028
John M. Scanlon http://orcid.org/0000-0001-8114-7546
Timothy L. McMurry http://orcid.org/0000-0001-5912-5092
Thomas Unger http://orcid.org/0009-0003-5428-3536

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